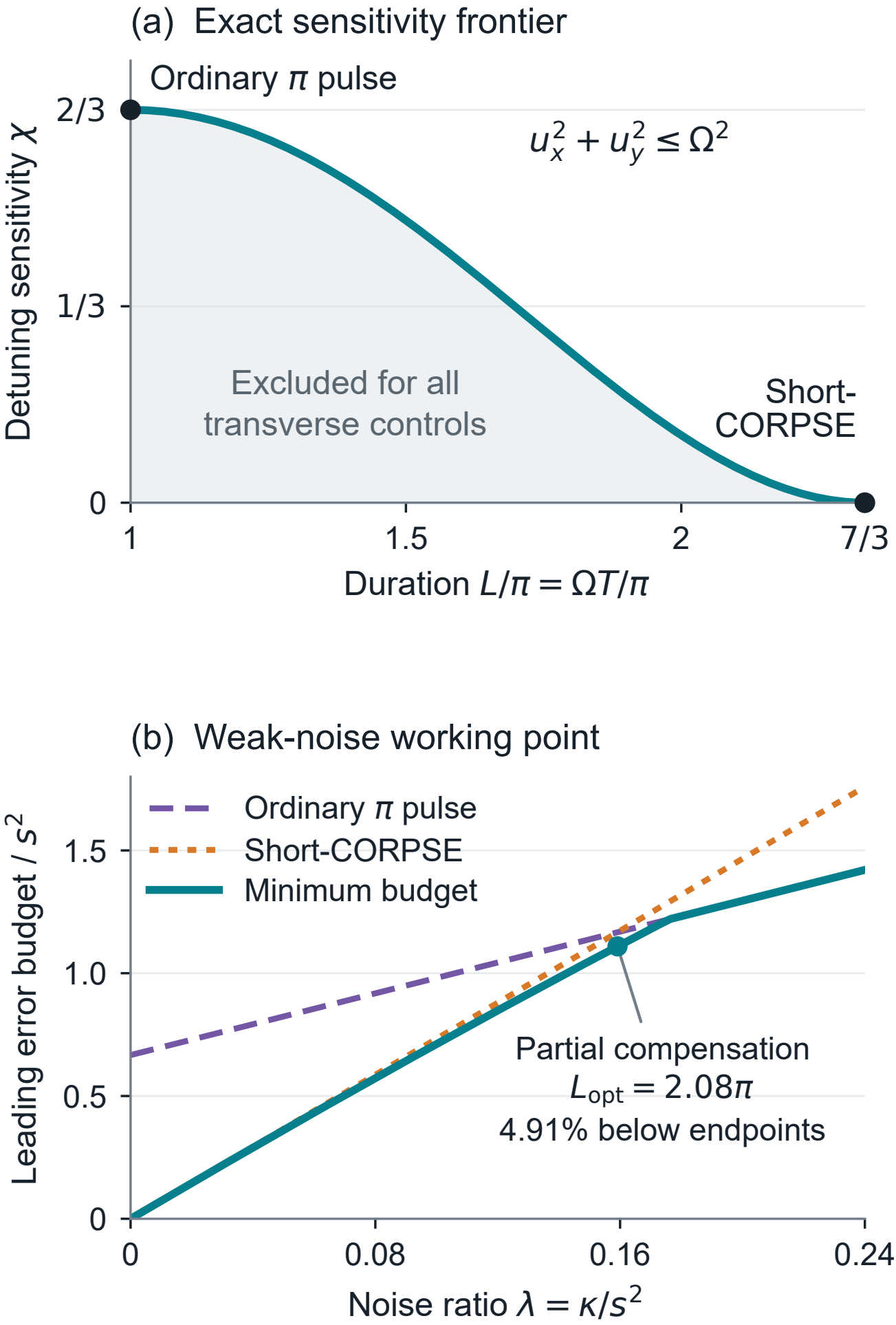
(a) Exact sensitivity frontier
Ordinary π pulse
$u_x^2 + u_y^2 \le \Omega^2$
Detuning sensitivity χ
2/3
1/3
0
Excluded for all
transverse controls
Short-
CORPSE
1
1.5
2
7/3
Duration $L/\pi = \Omega T/\pi$
(b) Weak-noise working point
Ordinary π pulse
Short-CORPSE
Minimum budget
Leading error budget / s^2
1.5
1.0
0.5
0
Partial compensation
$L_{opt} = 2.08\pi$
4.91% below endpoints
0
0.08
0.16
0.24
Noise ratio $\lambda = \kappa/s^2$